Phantasm parameter card

PARAM and two digits picks a parameter, VALUE and two digits sets it. Built from phantasm.bin sha256 2446ead910c5.

No.	Parameter	Range	CC	Values
1x	**Wave and sequence**			
10	Wavetable	0-n	-	the patch's tables, 0 first
11	Wave sequence	0-n	-	the patch's sequences, 0 first
12	Wave position	0-99	12	start, 0 first entry to 99 last, any table
13	Wave wrap	0-1	13	0 clamp, 1 wrap
14	Velocity > wave	0-99	14	
15	Key track > wave	0-99	15	
17	LFO > wave	0-99	17	
18	Sequence mode	0-4	18	0 off, 1 step, 2 per voice, 3 random, 4 LFO
19	Sequence reset gap	0-99	19	silence before restart, 0 never
2x	**Oscillators and pitch**			
20	Coarse A	38-62	20	semitones, 50 = 0
21	Fine A	29-71	21	1/256 octave (4.7 cents), 50 = 0
22	Coarse B	38-62	22	semitones, 50 = 0
23	Fine B	29-71	23	1/256 octave (4.7 cents), 50 = 0
24	Oscillator B level	0-99	24	0 = off; adds to A, so the voice gets louder
25	Bend range	0-12	25	whole semitones, full wheel; 2 is a whole tone, 12 an octave
26	Vintage	0-99	26	0 off; 1-99 more drift: each note-on gives every voice its own pitch and cutoff offset, up to +/-8 steps (+/-38 cents; +/-3.4 semitones of cutoff) at 99
27	Oscillator A level	0-99	27	99 = full, 0 = off
28	Oscillator B source	0-7	28	0 A's wave, 1 white noise, 2 pink, 3 scrape (tracking the key), 4 white, 5 pink, 6 scrape (fixed rate, C4), 7 sync (B is A's wave, hard-synced to A)
3x	**LFO, mod wheel and pressure**			
30	LFO rate	0-99	30	
31	LFO > pitch	0-99	31	
32	LFO shape	0-4	-	0 triangle, 1 ramp up, 2 ramp down, 3 square, 4 random
33	LFO sync	0-2	-	0 free-running, 1 restart at note-on, 2 MIDI clock (not built: runs as 0)
34	LFO > cutoff	0-99	34	
35	LFO delay	0-99	35	
36	Mod wheel depth	0-99	36	
37	Pressure depth	0-99	37	channel aftertouch, 0 = off
38	Mod wheel destination	0-1	-	0 LFO depth, 1 wave position
39	Pressure destination	0-2	39	0 LFO depth, 1 wave position, 2 cutoff
4x	**Filter envelope (ENV2)**			
40	ENV2 attack	0-99	40	
41	ENV2 peak	0-99	41	
42	ENV2 decay	0-99	42	
43	ENV2 sustain	0-99	43	
44	ENV2 release	0-99	44	
45	Velocity > ENV2	0-99	45	0 = depth as set (72); 99 scales it by velocity / 127
47	ENV2 key speed	1-99	47	50 = none; 99 halves decay and release per octave above key 60
5x	**Amp envelope (ENV1)**			
50	ENV1 attack	0-99	50	
51	ENV1 peak	0-99	51	
52	ENV1 decay	0-85	52	exponential
53	ENV1 sustain	0-99	53	
54	ENV1 release	0-85	54	exponential
57	ENV1 key speed	1-99	57	50 = none; as 47
6x	**Voice and keyboard modes**			
60	Voice mode	0-4	60	0 poly, 1 mono retrigger, 2 mono legato, 3 chord retrigger, 4 chord legato
61	Unison voices	1-8	61	per note; 1 = off
62	Unison spread	0-21	62	1/256 octave (4.7 cents) a step
63	Glide	0-99	63	0 off; 1-99 slower, an octave in 22 ms to 33 s
7x	**Filter and amp**			
70	Filter cutoff	0-99	70	
71	Resonance	0-99	71	high settings self-oscillate
72	ENV2 > cutoff	0-99	72	
73	Key track > cutoff	0-99	73	
74	Velocity > cutoff	0-99	74	
75	Amp level	0-99	75	lower it if chords or unison distort
76	Velocity > amp	0-99	76	
77	Wave > cutoff	1-99	77	bipolar, 50 = none: + opens the filter up the table
8x	**Modulation envelope (ENV3)**			
80	ENV3 attack	0-99	80	
81	ENV3 peak	0-99	81	
82	ENV3 decay	0-99	82	
83	ENV3 sustain	0-99	83	
84	ENV3 release	0-99	84	
85	ENV3 destination	0-5	85	0 pitch A+B, 1 pitch A, 2 pitch B, 3 resonance, 4 LFO depth, 5 wave position
86	ENV3 depth	0-99	86	bipolar, 50 = none
87	ENV3 key speed	1-99	87	50 = none; as 47
88	ENV3 loop	0-1	-	0 = off; 1 = restarts from its attack at the end of its decay while the key is held
9x	**Actions and instrument settings**			
93	Save user waves	action	-	the changed ones, at the second digit
94	Restart sequence	action	-	at the second digit
95	MIDI channel	1-16	-	numbered as a DAW numbers them
96	Omni	0-1	-	0 off, 1 on
97	Master tune	29-71	-	saved with the channel and omni; 50 = none, 4.7 cents a step
98	Transpose	38-62	-	saved with the channel and omni; 50 = none, semitones

CC: a MIDI controller of the same number edits the parameter; 0-127 scales onto its range,
and the display shows the edit. A dash means no CC: set it from the panel or by SysEx.
0-n: n is one less than the number of wavetables (10) or sequences (11) in the loaded patch.
action: happens when the second digit is keyed; there is no value to enter.